

1 Write a Python program to implement a menu-driven calculator.

Aim

To implement a menu-driven calculator in Python that allows the user to perform basic arithmetic operations (addition, subtraction, multiplication, division) repeatedly until they choose to exit.

Algorithm

1. Display a menu with options:
 1. Addition
 2. Subtraction
 3. Multiplication
 4. Division
 5. Exit
2. Accept the user's choice.
3. If the choice is between 1–4:
 - Ask the user to enter two numbers.
 - Perform the selected operation.
 - Display the result.
4. If the choice is 5:
 - Exit the program.
5. Repeat until the user chooses to exit.

CODE

```
def calculator():  
    while True:  
        print("\n--- Menu Driven Calculator ---")  
        print("1. Addition")  
        print("2. Subtraction")  
        print("3. Multiplication")  
        print("4. Division")
```

```
print("5. Exit")
```

```
choice = int(input("Enter your choice (1-5): "))
```

```
if choice == 5:
```

```
    print("Exiting the calculator. Goodbye!")
```

```
    break
```

```
if choice in [1, 2, 3, 4]:
```

```
    num1 = float(input("Enter first number: "))
```

```
    num2 = float(input("Enter second number: "))
```

```
    if choice == 1:
```

```
        print("Result:", num1 + num2)
```

```
    elif choice == 2:
```

```
        print("Result:", num1 - num2)
```

```
    elif choice == 3:
```

```
        print("Result:", num1 * num2)
```

```
    elif choice == 4:
```

```
        if num2 != 0:
```

```
            print("Result:", num1 / num2)
```

```
        else:
```

```
            print("Error: Division by zero is not allowed.")
```

```
    else:
```

```
        print("Invalid choice! Please enter a number between 1 and 5.")
```

```
# Run the calculator
```

```
calculator()
```

OUTPUT

--- Menu Driven Calculator ---

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Enter your choice (1-5): 1

Enter first number: 10

Enter second number: 5

Result: 15.0

--- Menu Driven Calculator ---

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Enter your choice (1-5): 4

Enter first number: 10

Enter second number: 0

Error: Division by zero is not allowed.

Enter your choice (1-5): 5

Exiting the calculator. Goodbye!

Conclusion

- The program successfully implements a menu-driven calculator.
- It allows repeated operations until the user chooses to exit.

- Handles invalid input and division by zero gracefully.

2 Write a program to print multiplication tables using loops.

Aim

To write a Python program that prints multiplication tables (e.g., from 1 to 10) using loops.

Algorithm

1. Ask the user for a number (or choose a range).
2. Use a loop to iterate from 1 to 10.
3. Multiply the given number by the loop counter.
4. Print the result in a formatted way.
5. Optionally, extend to print tables for multiple numbers.

CODE

```
# Program to print multiplication table using loops
```

```
# Single table
```

```
num = int(input("Enter a number to print its multiplication table: "))
```

```
print(f"\nMultiplication Table of {num}")
```

```
for i in range(1, 11):
```

```
    print(f"{num} x {i} = {num * i}")
```

```
# Multiple tables (1 to 10)
```

```
print("\nMultiplication Tables from 1 to 10")
```

```
for n in range(1, 11):
```

```
    print(f"\nTable of {n}")
```

```
        for i in range(1, 11):
```

```
print(f"{n} x {i} = {n * i}")
```

OUTPUT

Enter a number to print its multiplication table: 5

Multiplication Table of 5

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

$$5 \times 4 = 20$$

$$5 \times 5 = 25$$

$$5 \times 6 = 30$$

$$5 \times 7 = 35$$

$$5 \times 8 = 40$$

$$5 \times 9 = 45$$

$$5 \times 10 = 50$$

Multiplication Tables from 1 to 10

Table of 1

$$1 \times 1 = 1$$

$$1 \times 2 = 2$$

...

Table of 10

$$10 \times 1 = 10$$

$$10 \times 2 = 20$$

...

$$10 \times 10 = 100$$

Conclusion

- The program uses nested loops to generate multiplication tables.
- It can print a single table or a range of tables (1–10).
- This demonstrates how loops simplify repetitive tasks in Python.